

Week 7 Worksheet 1 Solutions

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Exercise 1. On midterm 1, you showed that if $T(a) = e^{-ipa/\hbar}$, then $T(a)\psi(x) = \psi(x - a)$. Generalize this result to three-dimensions.

The natural guess is that in 3-dimensions, if $\mathbf{a}$ is a 3-vector, then

$$T(\mathbf{a}) = e^{-i\mathbf{p}\cdot\mathbf{a}/\hbar},$$

and

$$T(\mathbf{a})\psi(\mathbf{x}) = \psi(\mathbf{x} - \mathbf{a}).$$

To show this, recall that since $[p_i, p_j] = 0$ for all $i, j \in \{x, y, z\}$,

$$T(\mathbf{a}) = e^{-ip_x a_x/\hbar} e^{-ip_y a_y/\hbar} e^{-ip_z a_z/\hbar} = T_x(a_x)T_y(a_y)T_z(a_z).$$

Now, the action of $T_z(a_z)$ on $\psi(\mathbf{x}) = \psi(x, y, z)$ is the same as in the 1-dimensional case, since as far as $T_z(a_z)$ is concerned, x and y are constants. Thus,

$$T_z(a_z)\psi(\mathbf{x}) = \psi(x, y, z - a_z).$$

This implies that

$$T_x(a_x)T_y(a_y)T_z(a_z)\psi(\mathbf{x}) = \psi(x - a_x, y - a_y, z - a_z) = \psi(\mathbf{x} - \mathbf{a}).$$

Exercise 2. For R a rotation matrix in 3-dimensions, let $U(R)$ be the operator defined by $U(R)|\mathbf{x}\rangle = |R\mathbf{x}\rangle$.

a) Show that $U = U(R)$ is unitary.

Hint: First, show that $\langle \mathbf{x}' | U^\dagger U | \mathbf{x} \rangle = \langle \mathbf{x}' | \mathbf{x} \rangle$ for all $\mathbf{x}$ and $\mathbf{x}'$. What else is left to show that $U(R)$ is unitary?

b) Show that $U(R_1)U(R_2) = U(R_1 R_2)$.

c) Show that $U(I) = \mathbb{1}$, where I is the identity (3×3) -matrix.

d) Show that $U(R^{-1}) = U(R)^{-1}$.

e) Show that if

$$|\psi'\rangle = U(R) |\psi\rangle ,$$

then

$$\psi'(\mathbf{x}) = \psi(R^{-1}\mathbf{x}).$$

Hint: Use a resolution of the identity.

Remark. This result is analogous to the result of the previous exercise,

$$T(\mathbf{a})\psi(\mathbf{x}) = \psi(\mathbf{x} - \mathbf{a}).$$

a) We need to show that $UU^\dagger = U^\dagger U = \mathbb{1}$. To do this, it is sufficient to identify all the matrix elements of the three operators. Following the hint, we check

$$\langle \mathbf{x}' | U^\dagger U | \mathbf{x} \rangle = \langle U \mathbf{x}' | U | \mathbf{x} \rangle = \langle R \mathbf{x}' | R \mathbf{x} \rangle = \delta(R(\mathbf{x} - \mathbf{x}')) = \delta(\mathbf{x} - \mathbf{x}') = \langle \mathbf{x}' | \mathbf{x} \rangle ,$$

since $R(\mathbf{x} - \mathbf{x}') = 0$ if and only if $\mathbf{x} - \mathbf{x}' = 0$. To figure out what happens for $UU^\dagger$, we need to figure out what $U^\dagger$ does to kets $|\mathbf{x}\rangle$. To see this, note that

$$\langle \mathbf{x}' | U^\dagger | \mathbf{x} \rangle = \langle U \mathbf{x}' | \mathbf{x} \rangle = \langle R \mathbf{x}' | \mathbf{x} \rangle = \delta(R \mathbf{x}' - \mathbf{x}).$$

Now, this δ -function will vanish unless $\mathbf{x}' = R^{-1}\mathbf{x}$, so it is the same as

$$\delta(R(\mathbf{x}' - R^{-1}\mathbf{x})) = \delta(\mathbf{x}' - R^{-1}\mathbf{x}) = \langle \mathbf{x}' | R^{-1}\mathbf{x} \rangle .$$

Since this is true for all $\mathbf{x}'$, this gives that

$$U^\dagger | \mathbf{x} \rangle = | R^{-1}\mathbf{x} \rangle .$$

Now, we can calculate

$$\langle \mathbf{x}' | UU^\dagger | \mathbf{x} \rangle = \langle R^{-1}\mathbf{x}' | R^{-1}\mathbf{x} \rangle = \langle \mathbf{x}' | \mathbf{x} \rangle$$

since we can replace $R \leftrightarrow R^{-1}$ in the argument for $U^\dagger U$.

b) It is sufficient to check that the two operators in question match up on all kets $|\mathbf{x}\rangle$. We check

$$U(R_1)U(R_2) |\mathbf{x}\rangle = U(R_1) |R_2\mathbf{x}\rangle = |R_1 R_2\mathbf{x}\rangle = U(R_1 R_2)\mathbf{x}.$$

c) We check

$$U(I) |\mathbf{x}\rangle = |\mathbf{x}\rangle = \mathbb{1} |\mathbf{x}\rangle .$$

d) This follows from the answer to part (a), where we found $U^\dagger = U^{-1} = U(R^{-1})$.

e) Following the hint, we use a resolution of the identity to compute

$$\begin{aligned} |\psi'\rangle &= \int d^3x' U(R) |\mathbf{x}'\rangle \langle \mathbf{x}'|\psi\rangle \\ &= \int d^3x' |R\mathbf{x}'\rangle \langle \mathbf{x}'|\psi\rangle. \end{aligned}$$

Since we're interested in $\psi'(\mathbf{x}) = \langle \mathbf{x}|\psi'\rangle$, we should apply $\langle \mathbf{x}|$ to the above formula. This gives

$$\begin{aligned} \psi'(\mathbf{x}) &= \int d^3x' \langle \mathbf{x}|R\mathbf{x}'\rangle \psi(\mathbf{x}') \\ &= \int d^3x' \delta(\mathbf{x} - R\mathbf{x}') \psi(\mathbf{x}') \\ &= \psi(R^{-1}\mathbf{x}), \end{aligned}$$

since $\mathbf{x} = R\mathbf{x}'$ implies $\mathbf{x}' = R^{-1}\mathbf{x}$.